

4. A Welsh farmer decides to purchase a wind generator to provide some electrical energy for his farm.
A selection of wind generators is available.

	Wind generator 1	Wind generator 2	Wind generator 3
Cost to buy and install (£)	1 800	3 000	2 925
Expected lifetime (years)	15	15	15
Estimated number of units produced per year (kWh)	2 000	4 000	3 000
Cost per unit (£)	0.15	0.15	0.15
Estimated saving per year (£)		600	450
Estimated payback time (years)		5.0	6.5

- (a) (i) Use the equation:

$$\text{estimated saving per year} = \text{units produced per year} \times \text{cost per unit}$$

and data from the table to calculate the estimated saving per year if **wind generator 1** was installed.

[1]

Estimated saving per year = £

- (ii) Use the equation:

$$\text{estimated payback time} = \frac{\text{cost to buy and install}}{\text{estimated saving per year}}$$

and data from the table to calculate the estimated payback time for **wind generator 1**.

[2]

Estimated payback time = years



- (b) (i) A salesperson suggests to the farmer that **wind generator 3** is better than **wind generator 2** because it is cheaper.
Explain, using information from the table, if the salesperson's suggestion is correct over a 5-year timescale. [2]

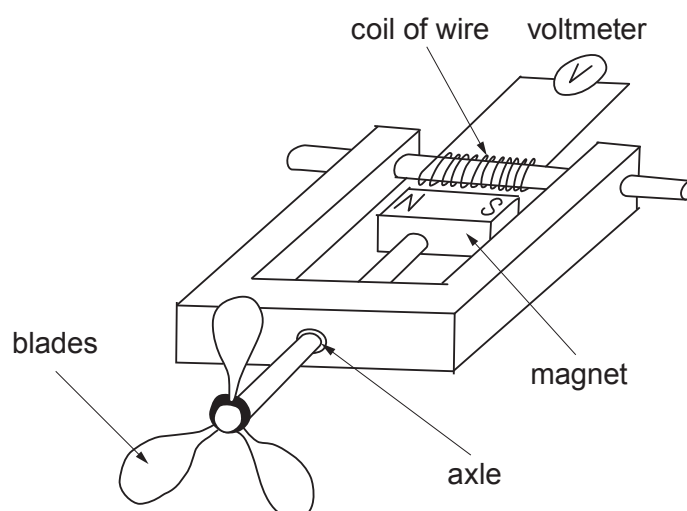
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- (ii) Give **two** reasons why the payback time can only be an estimate. [2]
1.
2.

- (c) A wind generator only operates in certain wind speeds.
The diagram below shows a student's design for measuring wind speed.



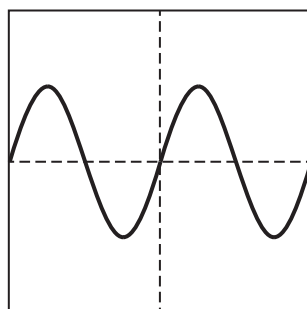
- (i) When the wind blows a reading on the voltmeter is obtained.
The following sentences describe the different stages.
- A** The magnetic field lines cut through the coil of wire.
 - B** The blades rotate.
 - C** The magnet rotates.
 - D** This induces a voltage in the coil of wire.

Put the letters **A**, **B** and **C** in the boxes to correctly describe the process of inducing a voltage in the coil of wire. The final box has already been completed, [2]

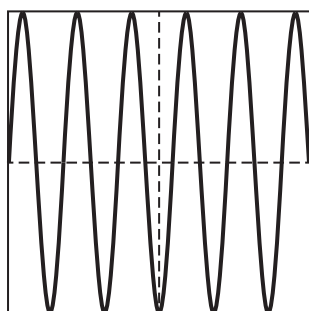
→ → → **D**



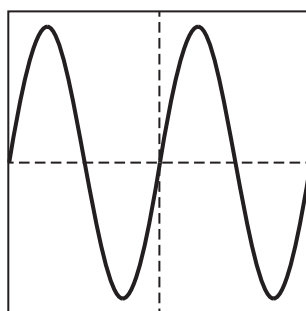
- (ii) When the wind speed is **2 m/s** the alternating voltage produced is shown below.



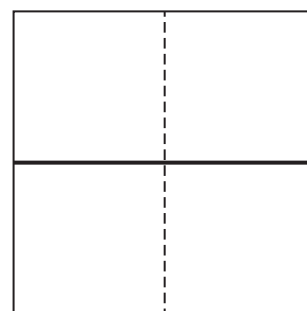
The following traces **X**, **Y** and **Z** are produced during different wind conditions.



Trace **X**



Trace **Y**



Trace **Z**

- I. State which trace (**X**, **Y** or **Z**) is produced when the wind speed is **faster than 2 m/s**.

Trace [1]

- II. State which trace (**X**, **Y** or **Z**) is produced when **the wind stops**.

Trace [1]

- (d) When the student's design is built it is found that the voltmeter is not sensitive enough to measure a wind speed less than 0.5 m/s.

- (i) Suggest **one** change that can be made to the **magnet** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

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- (ii) Suggest **one** change that can be made to the **coil of wire** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

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